

"What is wrong with my crop and how can I treat it?"

Vision Module

ResNet-50
2048-dim features

EfficientNet-B3
1536-dim features

DenseNet-121
1024-dim features

Soft Vote
Softmax avg

FAISS
4608-dim NN

(low-confidence fallback)

Confidence Threshold 80%

Knowledge Retrieval

AgroBench KB
38 disease entries

Kisan Call Center
178 K advisories

Context Retrieval
• Location
• Weather Context

LLaVA-Next 7B

Stage 1
Visual Grounding

Stage 2
Agronomic Reasoning

Advisory Validation
Knowledge Consistency Check

Multilingual Output
11 languages